

- **Energy Consumption Forecasting:** Predicting electricity usage based on historical consumption patterns.
- **Life Expectancy Prediction:** Estimating lifespan based on health, demographic, and lifestyle factors.

b. Regression Models

- **Linear Regression**
 - **Why:** Simple, transparent, computationally efficient.
 - **Best for:** Predicting continuous outcomes where relationship is linear or nearly linear (e.g., sales forecasting).
- **Random Forest Regressor**
 - **Why:** Handles nonlinear relationships, reduces variance, resistant to outliers.
 - **Best for:** Complex, nonlinear data (e.g., house price predictions, energy consumption forecasting).
- **Gradient Boosted Trees (XGBoost, LightGBM)**
 - **Why:** High predictive accuracy, handles large datasets efficiently.
 - **Best for:** Structured/tabular data, often wins competitions (e.g., stock market prediction, customer lifetime value estimation).
- **Support Vector Regression (SVR)**
 - **Why:** Effective in handling high-dimensional spaces, robust against overfitting.
 - **Best for:** Smaller datasets with clear patterns (e.g., predicting performance metrics).
- **Neural Networks (Deep Regression)**
 - **Why:** Excellent at modeling very complex, nonlinear relationships.
 - **Best for:** Large datasets, particularly time series or complex feature sets (e.g., predicting stock prices, weather forecasting).

Example Classification: Assigning labels (categories) to inputs.

Example: Spam detection in emails.